

Integer Linear Programming

Binary

Further Problems

$$x_i \in \mathbb{Z}$$

$$x_i \geq 0$$

$$y_i \leq 1$$

Blank Slide

Minimum Set Cover

General Set Covering Problem

minimise

$$\sum_{s \in S} x_s$$

subject to:

$$\sum_{s: e \in S} x_s \geq 1, \quad \forall e \in U$$

$$x_s \in \{0,1\}$$

S – All possible sets to choose from

$s \in S$ – A given set s

x_s – Set s chosen or not

N – number of sets $|S|$

U = Set of all elements contained in S

General Set Covering Problem

minimise

$$\sum_{s \in S} x_s$$

minimise no of sets picked

subject to:

$$\sum_{s: e \in s} x_s \geq 1, \quad \forall e \in U$$

$x_s \in \{0,1\}$

S s.t e a member of s

$$x_a \geq 1$$

$$x_a + x_b \geq 1$$

Example

Set

$$a = \{1,2,5\}, \quad b = \{2,3\},$$

$$c = \{4,5\}, \quad d = \{3,4\}$$

Set

$$S = \{a, b, c, d\}$$

$$U = \{1,2,3,4,5\}$$

all elements

$$N = 4$$

Pick the minimum number of sets to get all elements in U

Answer by inspection?

General Set Covering Problem

minimise

$$\sum_{s \in S} x_s$$

subject to:

$$\sum_{s: e \in S} x_s \geq 1, \quad \forall e \in U$$

$$x_s \in \{0,1\}$$

$$a = \{1,2,5\}, \quad b = \{2,3\}, \quad c = \{4,5\}, \quad d = \{3,4\}$$

$$S = \{a, b, c, d\}$$

$$U = \{1,2,3,4,5\}$$

$$N = 4$$



minimise

subject to:

$$x_a + x_b + x_c + x_d$$

$$x_a \geq 1 \quad (\text{Element 1})$$

$$x_a + x_b \geq 1 \quad (\text{Element 2})$$

$$x_b + x_d \geq 1 \quad (\text{Element 3})$$

$$x_c + x_d \geq 1 \quad (\text{Element 4})$$

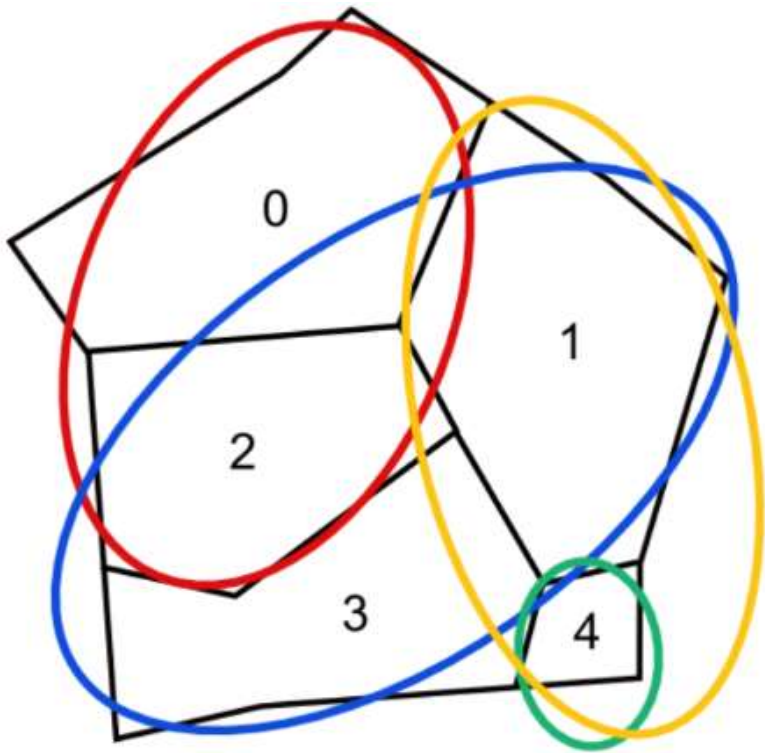
$$x_a + x_c \geq 1 \quad (\text{Element 5})$$

$$x_s \in \{0,1\}$$

Minimum set cover 2

$$\{a, d\}$$

Costed Fire Station Example



S – All possible firestations to choose from

$s \in S$ – A given firestation s

x_s – firestation s chosen or not

c_s – cost of running firestation s

U = Set of all cities contained in S

$$S = \{a, b, c, d\}$$

$$U = \{0, 1, 2, 3, 4\}$$

$$N = 4$$

$$a = \{0, 2\} \quad \leftarrow \text{covers cities}$$

$$b = \{1, 2, 3\}$$

$$c = \{1, 4\}$$

$$d = \{4\}$$

$$c_a = 12$$

$$c_b = 7$$

$$c_c = 10$$

$$c_d = 5$$

Costed Fire Station Example

S – All possible firestations to choose from

$s_i \in S$ – A given firestation s_i

x_i – firestation s_i chosen or not

c_i – cost of running firestation i

U = Set of all cities contained in S

$$S = \{a, b, c, d\}$$

$$U = \{0, 1, 2, 3, 4\}$$

$$s_a = \{0, 2\}$$

$$s_b = \{1, 2, 3\}$$

$$s_c = \{1, 4\}$$

$$s_d = \{4\}$$

$$c_a = 12$$

$$c_b = 7$$

$$c_c = 10$$

$$c_d = 5$$

minimise

subject to:

$$\sum_{s \in S} c_s x_s$$

Minimise cost of picking firestation

$$\sum_{s: e \in s} x_s \geq 1, \quad \forall e \in U$$

$$x_s \in \{0, 1\}$$

↓

minimise

subject to:

$$\rightarrow c_a x_a + c_b x_b + c_c x_c + c_d x_d$$

$$x_a \geq 1 \quad (\text{City 0 covered}) \checkmark$$

$$x_b + x_c \geq 1 \quad (\text{City 1 covered})$$

$$x_a + x_b \geq 1 \quad (\text{City 2 covered})$$

$$x_b \geq 1 \quad (\text{City 3 covered})$$

$$x_c + x_d \geq 1 \quad (\text{City 4 covered})$$

$$x_s \in \{0, 1\}$$

Costed Fire Station Example

S – All possible firestations to choose from

$s_i \in S$ – A given firestation s_i

x_i – firestation s_i chosen or not

c_i – cost of running firestation i

U = Set of all cities contained in S

$$S = \{a, b, c, d\}$$

$$U = \{0, 1, 2, 3, 4\}$$

$$s_a = \{0, 2\}$$

$$s_b = \{1, 2, 3\}$$

$$s_c = \{1, 4\}$$

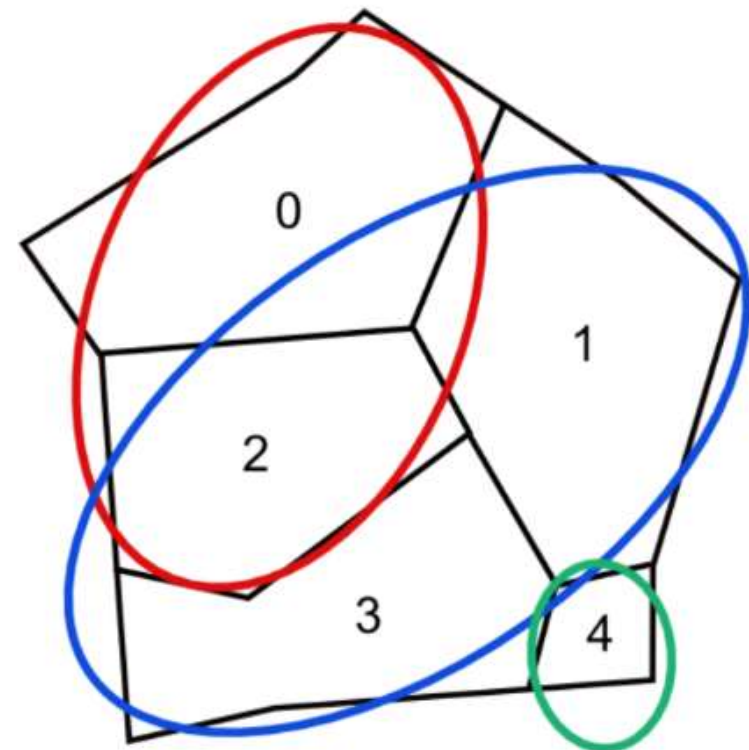
$$s_d = \{4\}$$

$$c_a = 12$$

$$c_b = 7$$

$$c_c = 10$$

$$c_d = 5$$

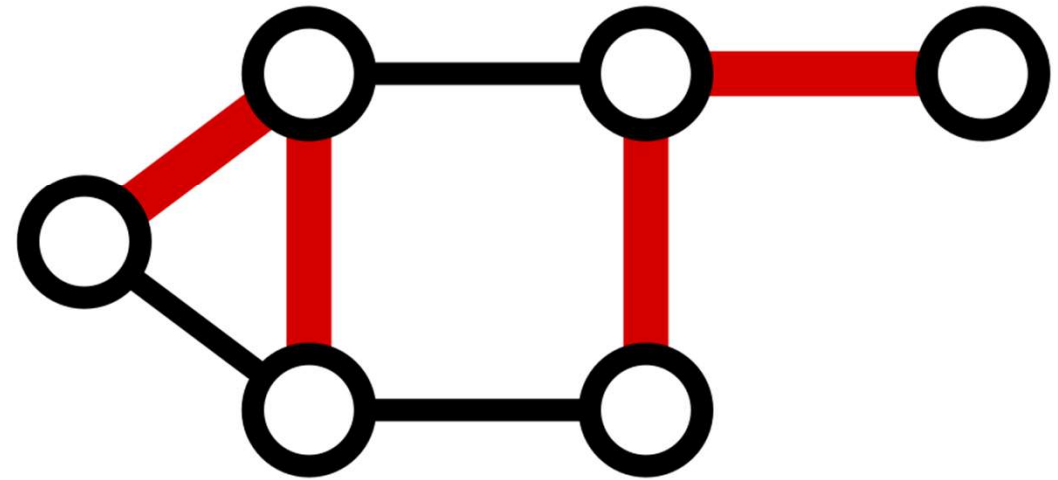
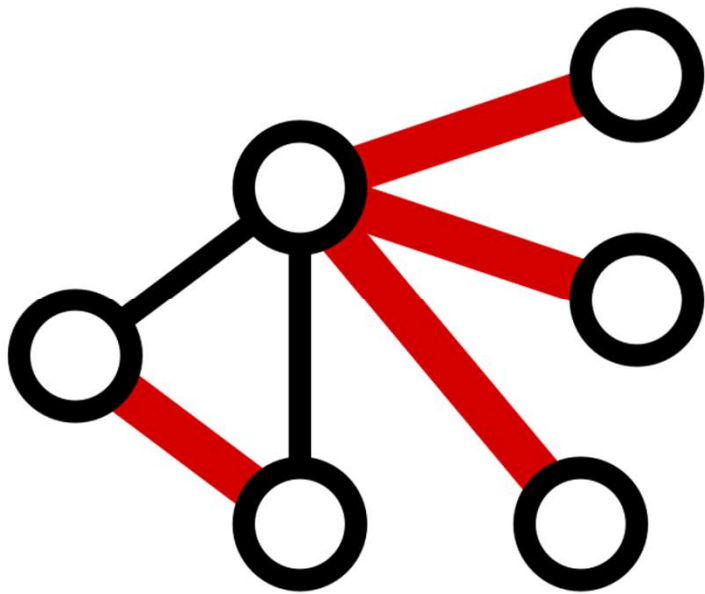


$$\text{cost} = 12 + 7 + 5 = 24$$

Edge Covering

Find a set of edges that cover the vertices...

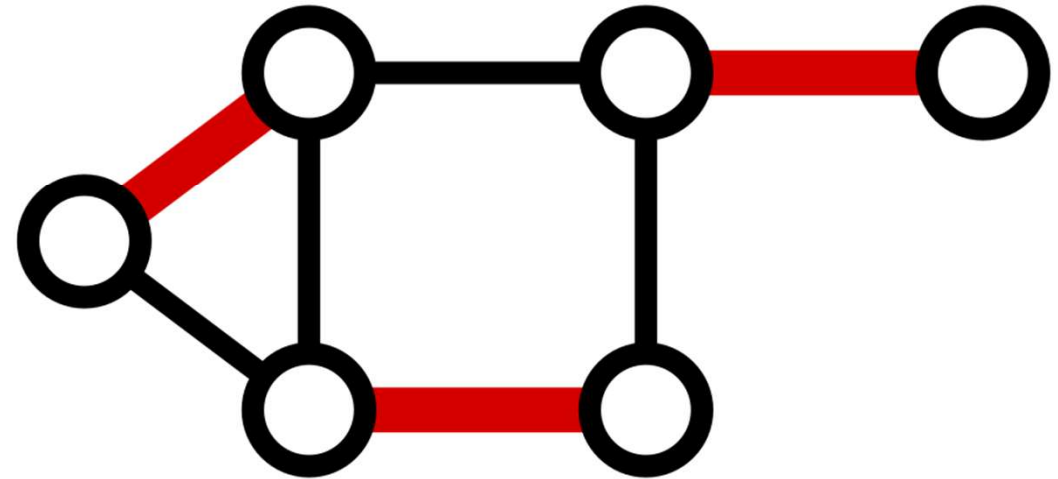
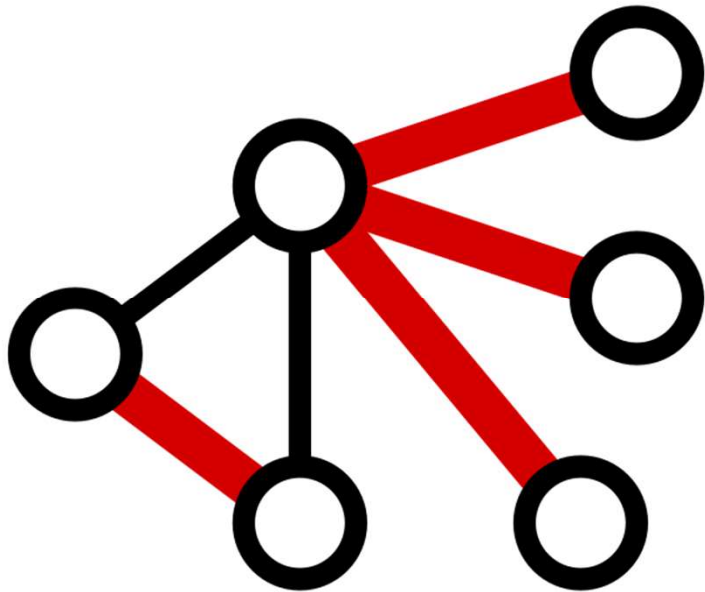
e.g.



Minimum Edge Covering

Find a the minimum number of edges that cover the vertices...

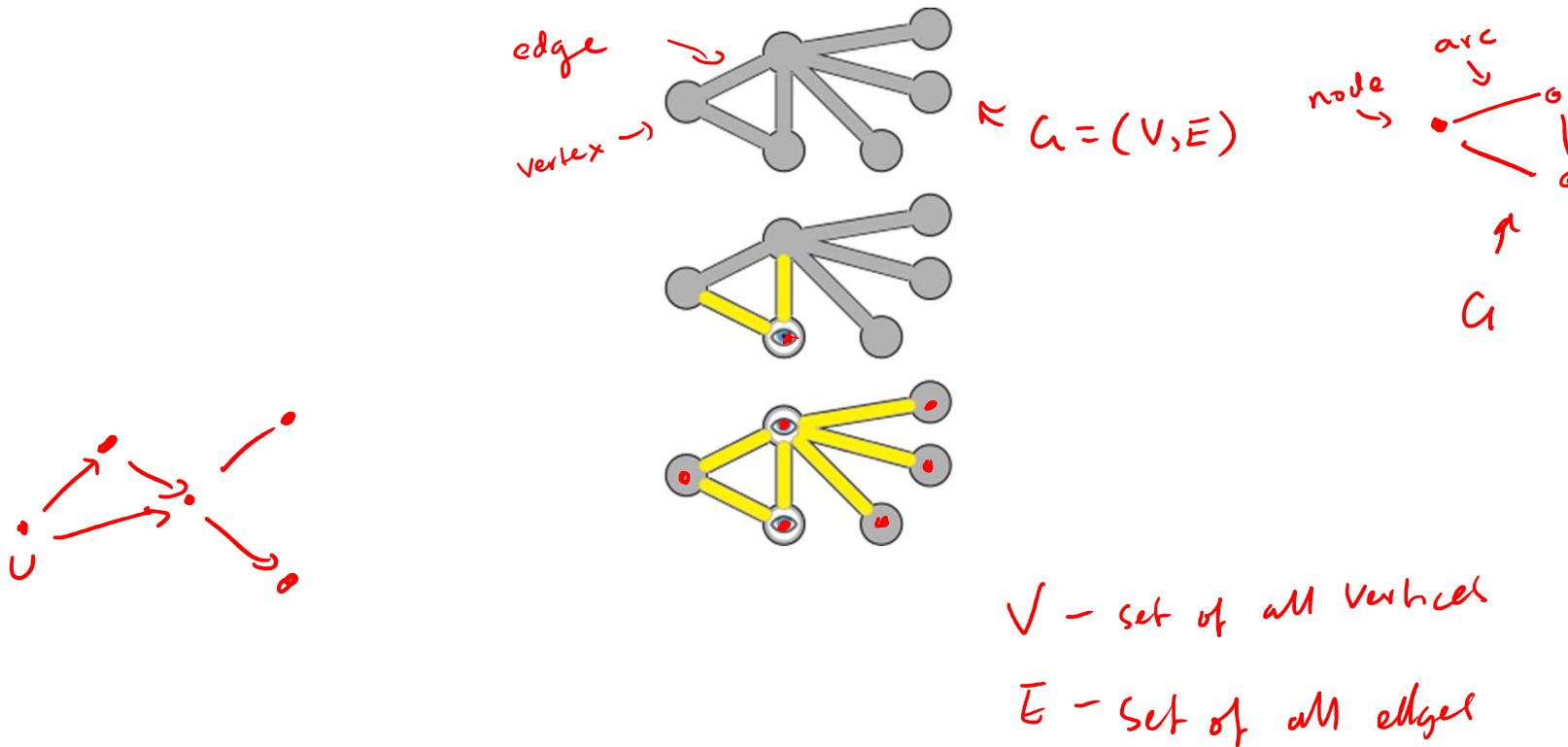
e.g.



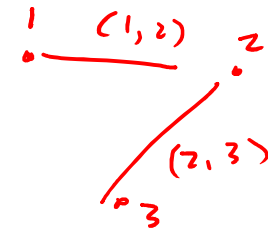
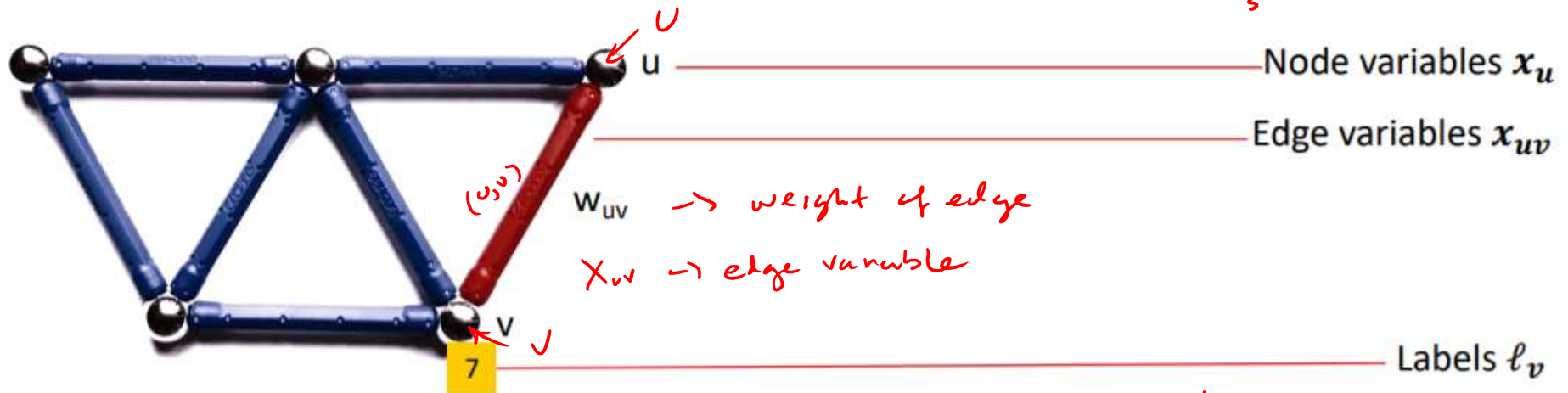
Minimum Vertex Cover

Vertex Cover

A set of vertices that includes at least one endpoint of every edge



Minimum Vertex Cover



Example 2: Min. vertex cover

$\min \sum_{u \in V} x_u$

s.t. $\forall \{u, v\} \in E: x_u + x_v \geq 1$

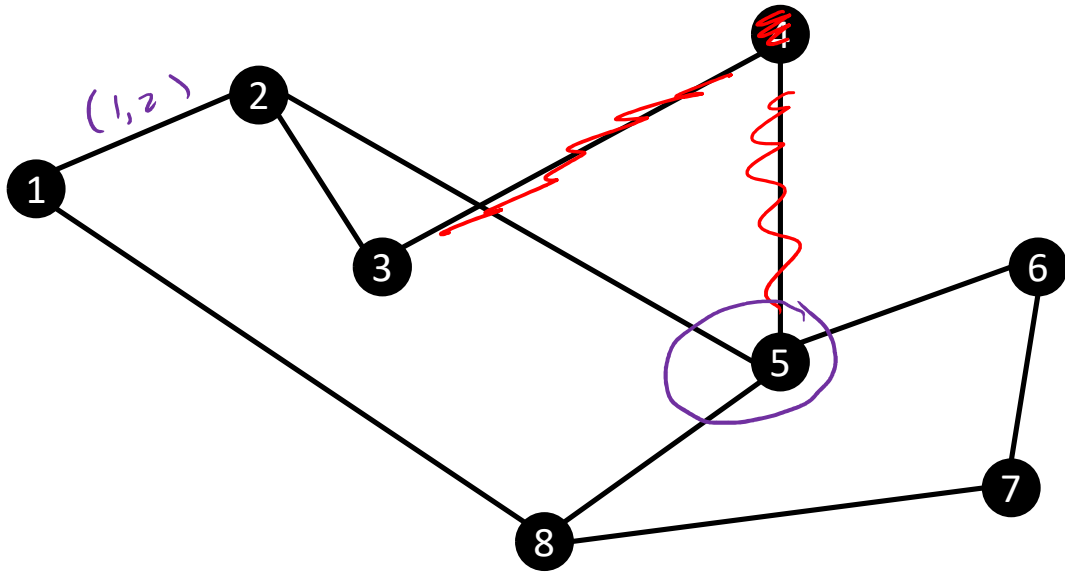
Handwritten red notes: "Pick min no of vertices", "Pick vertex 1 or vertex 2", $x_1 + x_2 \geq 1$

Handwritten red notes: $x_1 + x_2 \geq 1$

Example – Installing Security Cameras

- You wish to install security cameras in a school to cover every corridor.
- Obviously you would like to install as few as needed
- The following example is from <https://brilliant.org/wiki/vertex-cover/>

Example – Installing Security Cameras



minimise

$$\sum_{u \in V} x_u = x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 + x_8$$

subject to:

$$x_1 + x_2 \geq 1 \quad \leftarrow$$

$$x_1 + x_8 \geq 1$$

$$x_2 + x_3 \geq 1$$

$$x_2 + x_5 \geq 1$$

$$x_3 + x_4 \geq 1$$

$$x_4 + x_5 \geq 1$$

$$x_5 + x_6 \geq 1$$

$$x_5 + x_8 \geq 1$$

$$x_6 + x_7 \geq 1$$

$$x_7 + x_8 \geq 1$$

$$x_u \in \{0,1\}$$

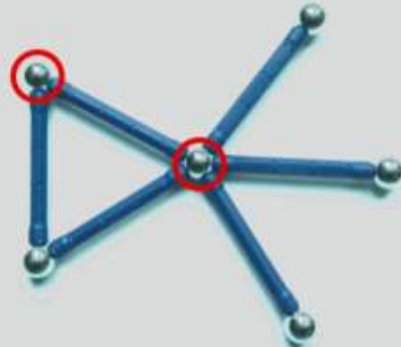


Example 2: **Min. vertex cover**

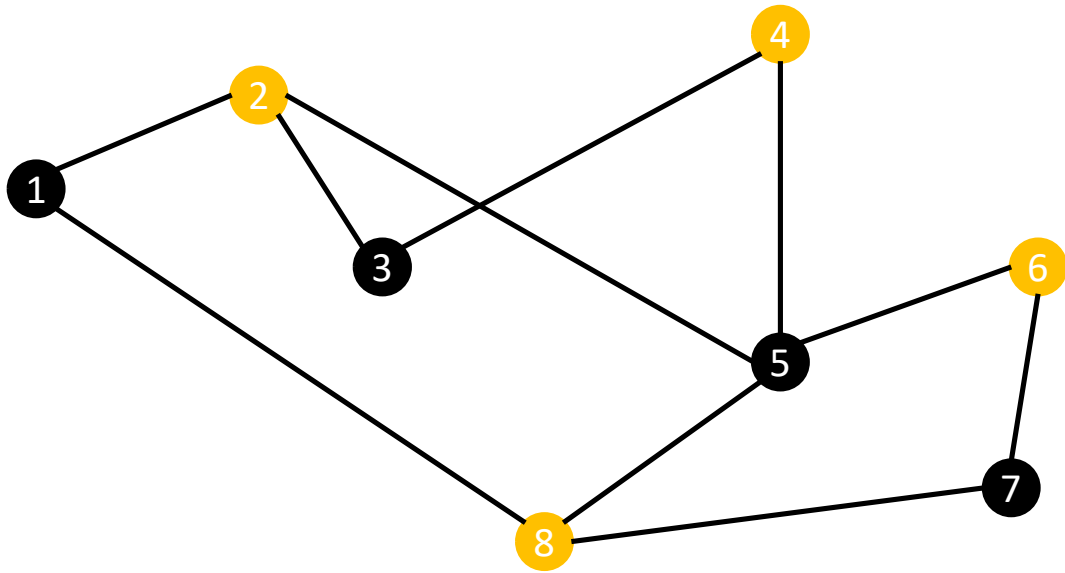
$$\min \sum_{u \in V} x_u$$

s.t.

$$\forall \{u, v\} \in E: x_u + x_v \geq 1$$



Example – Installing Security Cameras



MATLAB Solution

minimise

$$\sum_{u \in V} x_u = x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 + x_8$$

subject to:

$$x_1 + x_2 \geq 1$$

$$x_1 + x_8 \geq 1$$

$$x_2 + x_3 \geq 1$$

$$x_2 + x_5 \geq 1$$

$$x_3 + x_4 \geq 1$$

$$x_4 + x_5 \geq 1$$

$$x_5 + x_6 \geq 1$$

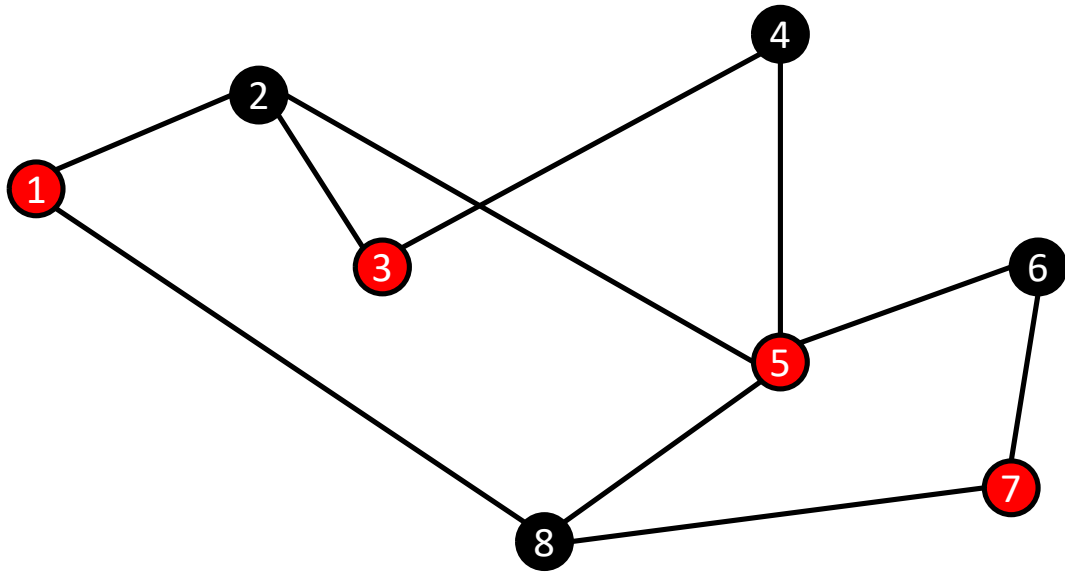
$$x_5 + x_8 \geq 1$$

$$x_6 + x_7 \geq 1$$

$$x_7 + x_8 \geq 1$$

$$x_u \in \{0,1\}$$

Example – Installing Security Cameras



Another possibility

minimise

$$\sum_{u \in V} x_u = x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 + x_8$$

subject to:

$$x_1 + x_2 \geq 1$$

$$x_1 + x_8 \geq 1$$

$$x_2 + x_3 \geq 1$$

$$x_2 + x_5 \geq 1$$

$$x_3 + x_4 \geq 1$$

$$x_4 + x_5 \geq 1$$

$$x_5 + x_6 \geq 1$$

$$x_5 + x_8 \geq 1$$

$$x_6 + x_7 \geq 1$$

$$x_7 + x_8 \geq 1$$

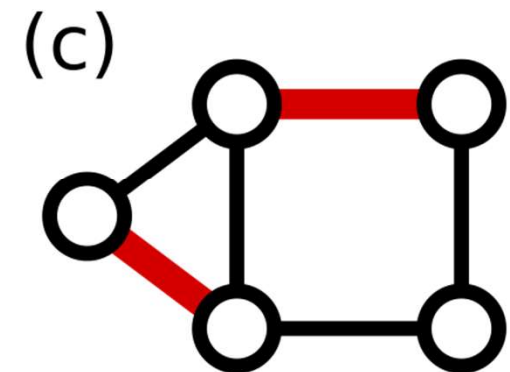
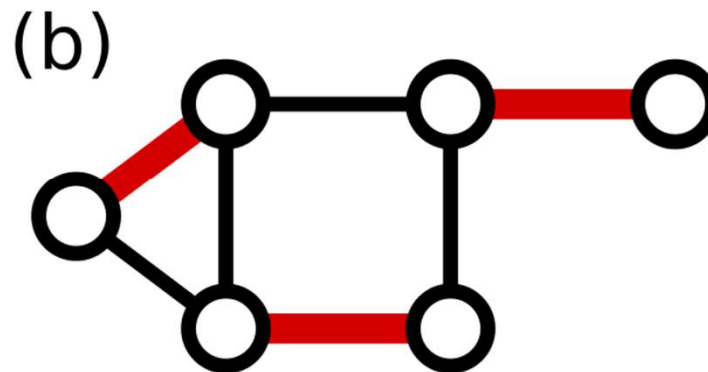
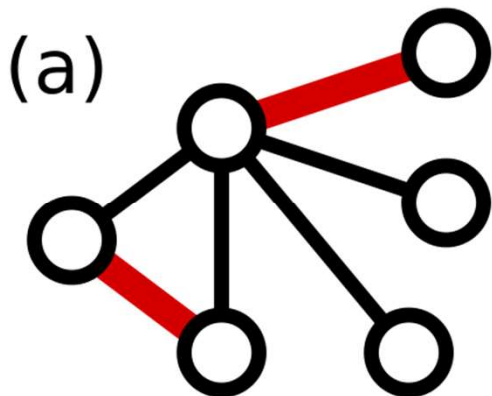
$$x_u \in \{0,1\}$$

Maximum Weight Matching and Assignment Problem

Maximum Matching

- Matching – Set of edges without common vertices
- Maximum Matching – Matching with the largest number of edges

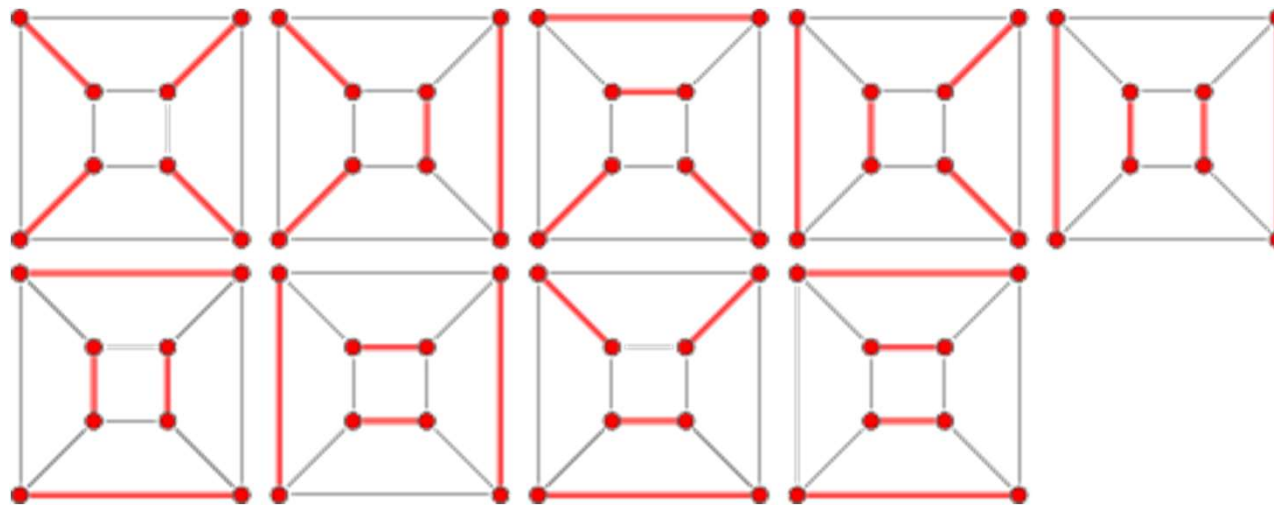
Examples of Maximum Matchings



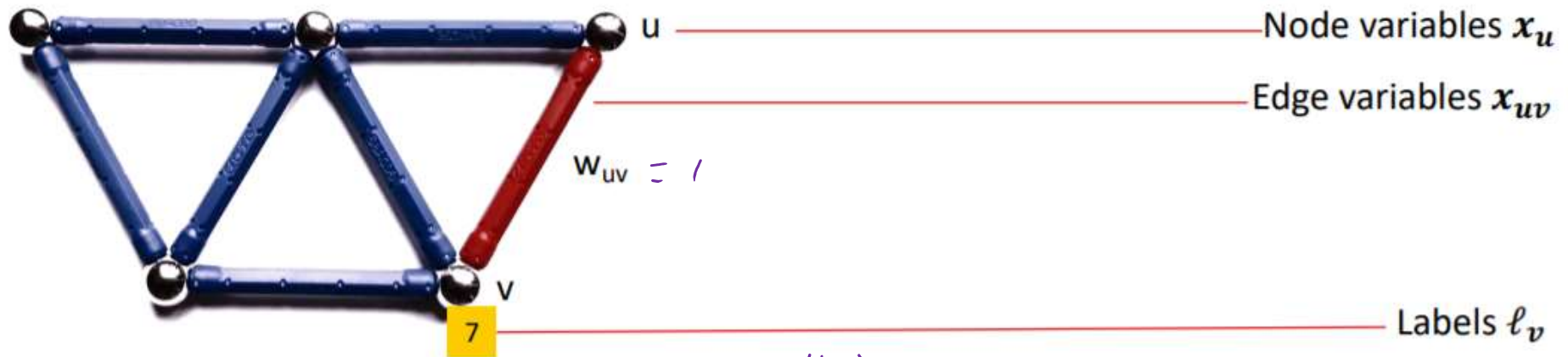
Perfect Matching

- Matching – Set of edges without common vertices
- Perfect Matching – Matching that includes every vertex

Examples



Maximum Weight Matching

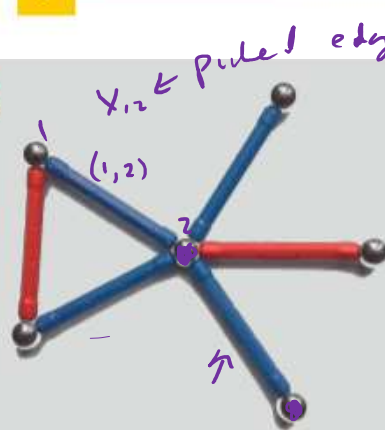


Example 1: Max. weight matching

$$\max \sum_{\{u,v\} \in E} w_{uv} \cdot x_{uv}$$

s.t. *for every edge*

$$\forall u \in V: \sum_{\{u,v\} \in E} x_{uv} \leq 1$$



- Select a set of edges without common vertices
- Maximise the weight of the edges

Balanced Assignment Problem

- Assign a worker to each job – minimise completion time

	Job 1	Job 2	Job 3	Job 4
Worker 1	9	2	7	8
Worker 2	6	4	3	7
Worker 3	5	8	1	8
Worker 4	7	6	9	4

This is a bi-partite graph and requires a perfect matching...

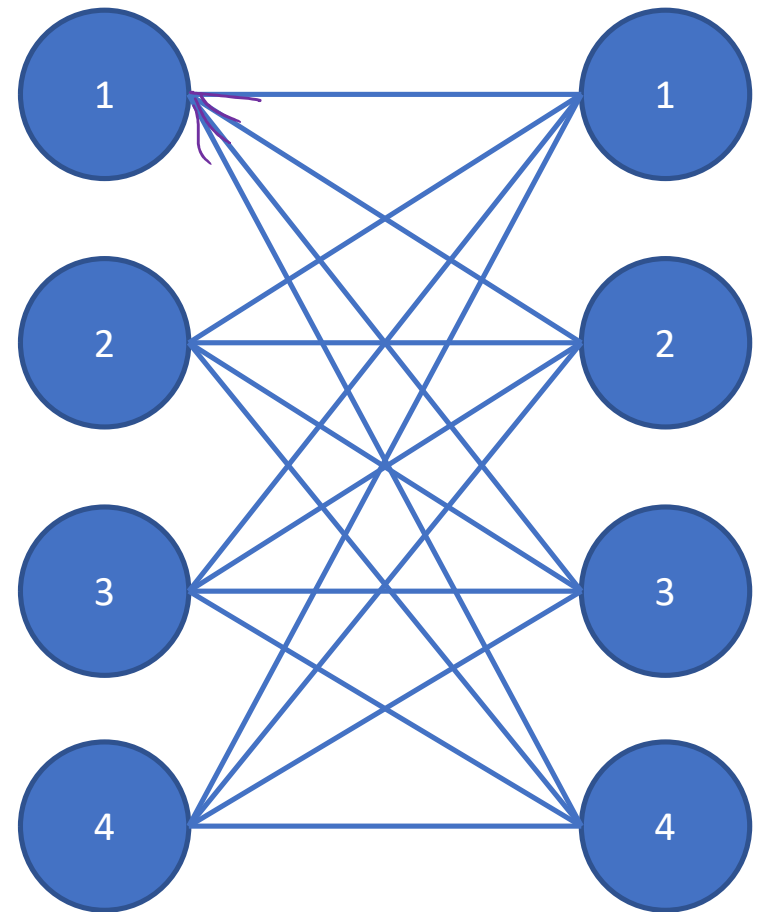
How many solutions??

4 3 2 1

4!

In general $n!$

Sensitivity: Internal



Balanced Assignment Problem

- Example Solution

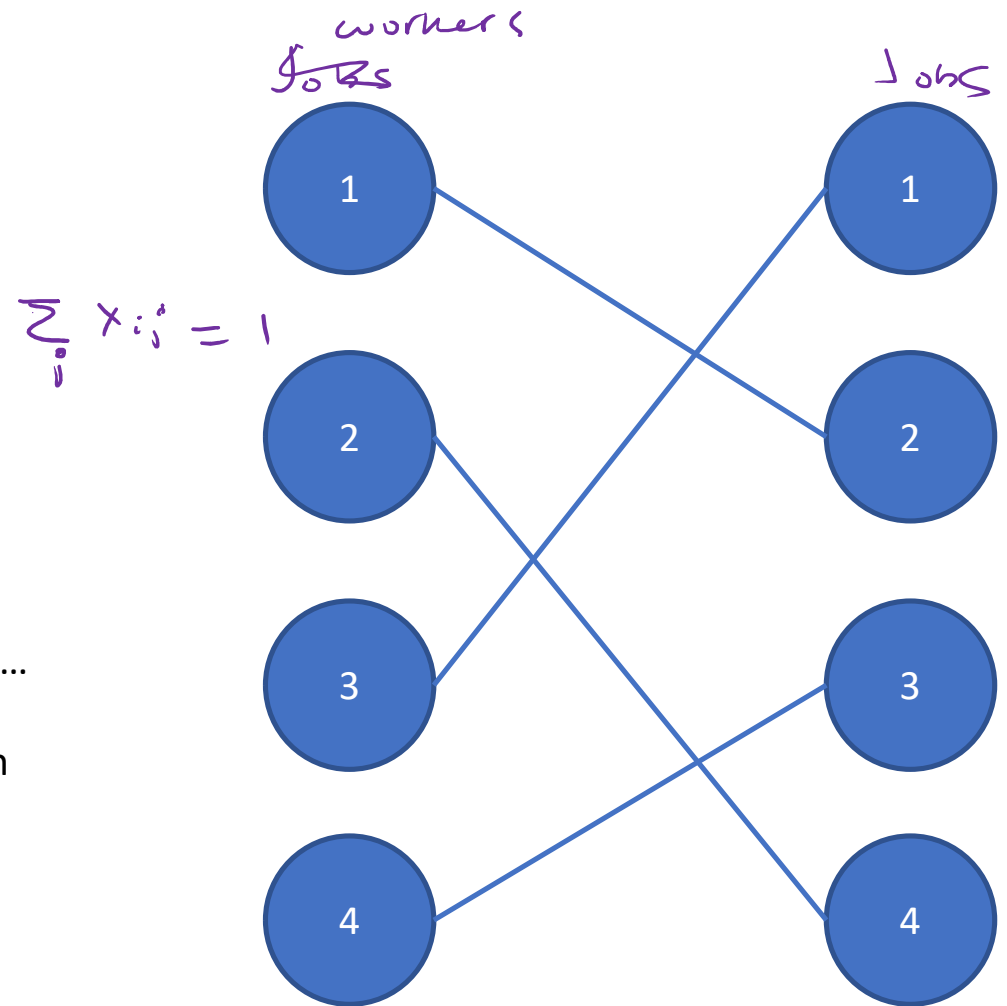
	Job 1	Job 2	Job 3	Job 4
Worker 1	9	2	7	8
Worker 2	6	4	3	7
Worker 3	5	8	1	8
Worker 4	7	6	9	4

$$\sum_j x_{ij} = 1$$

This is a bi-partite graph and requires a perfect matching...

Here every vertex is included and no edge has a common vertex

Cost is $2 + 7 + 5 + 9 = 23$



Balanced Assignment Problem – Model

- Pick 1 Per Row and 1 Per column

	Job 1	Job 2	Job 3	Job 4
Worker 1	9	2	7	8
Worker 2	6	4	3	7
Worker 3	5	8	1	8
Worker 4	7	6	9	4

W - Set of Workers

J - Set of Jobs

x_{uv} - Worker u completes job v

c_{uv} - Completion time for worker u completing job v

minimise

$$\sum_{u \in W} \sum_{v \in J} c_{uv} x_{uv} = c_{11}x_{11} + c_{12}x_{12} + \dots + c_{44}x_{44}$$

subject to:

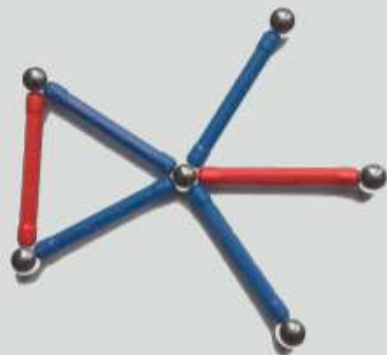
$$\sum_{v \in J} x_{uv} = 1, \quad \forall u \in W$$

$$\sum_{u \in W} x_{uv} = 1, \quad \forall v \in J$$

$$x_{uv} \in \{0,1\}$$

Example 1: Max. weight matching

$$\begin{aligned} \max \quad & \sum_{\{u,v\} \in E} w_{uv} \cdot x_{uv} \\ \text{s.t.} \quad & \\ & \forall u \in V: \sum_{\{u,v\} \in E} x_{uv} \leq 1 \end{aligned}$$



Balanced Assignment Problem

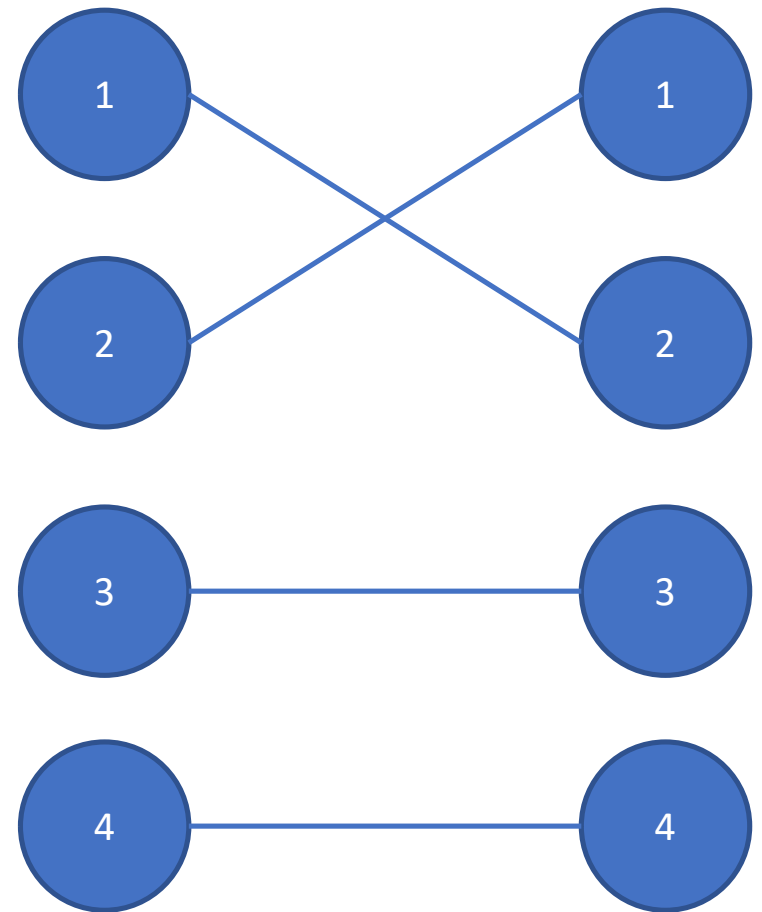
- Optimal Solution

	Job 1	Job 2	Job 3	Job 4
Worker 1	9	2	7	8
Worker 2	6	4	3	7
Worker 3	5	8	1	8
Worker 4	7	6	9	4

This is a bi-partite graph and requires a perfect matching...

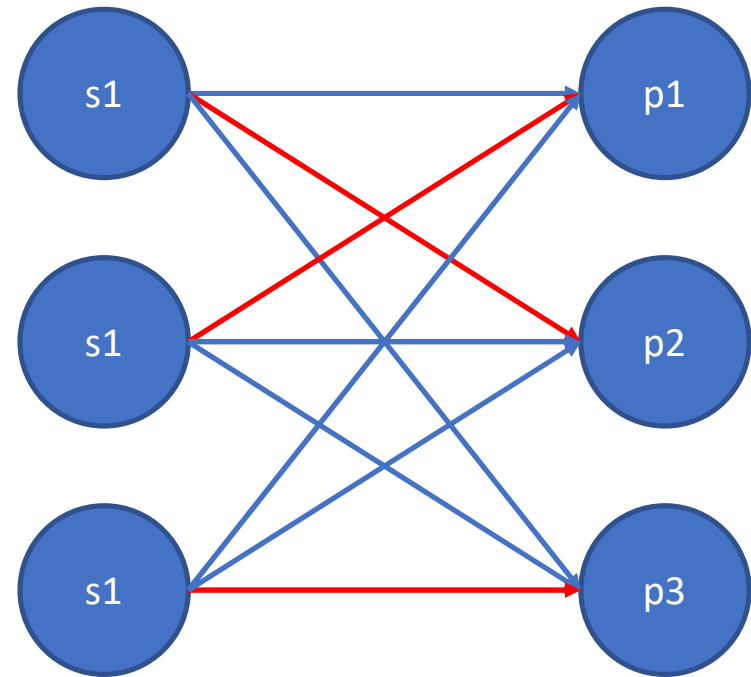
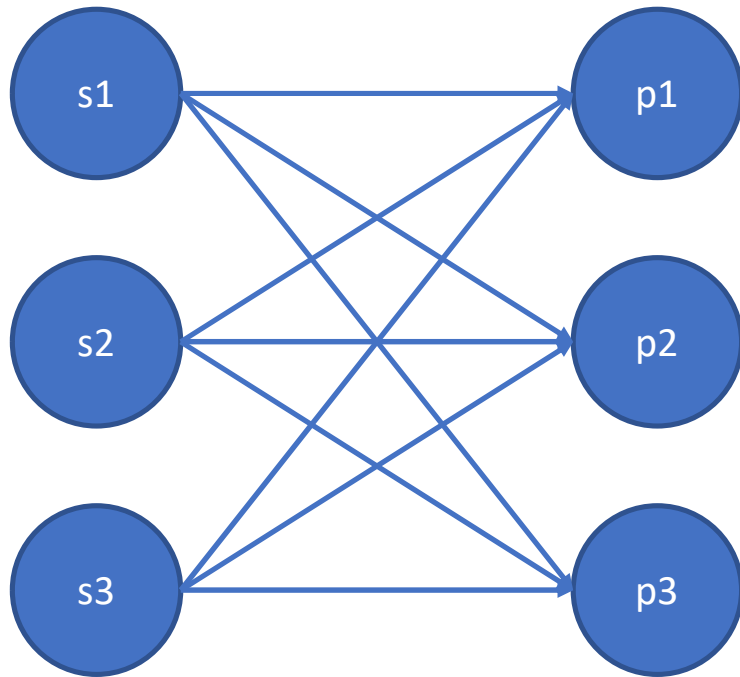
Here every vertex is included and no edge has a common vertex

Cost is $2 + 6 + 1 + 4 = 13$



Assignment Problem

Assign 3 students to the 3 different dissertation projects.



Tutorial

1. Study these models and understand the Python code. It will be useful for coming lectures.
2. Code the costed fire station in Pyomo using an incidence matrix (matrix of 0s and 1s). Think about how each element (city) is covered by each fire station.
3. Look at the examples on Gurobi - <https://www.gurobi.com/resource/modeling-examples-using-the-gurobi-python-api-in-jupyter-notebook/>
4. Pick an example and try coding it in Pyomo
5. Try coding these yourself.

Look up other problems such as.

- Maximum Set Cover
- Maximum Edge Cover
- Maximum Independent Set